

RISHI GARG

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[LinkedIn](#) | [GitHub](#) | [Leetcode](#) | [Portfolio](#)

EDUCATION

B.TECH(Software Engineering)	2025-2029	Delhi Technological University, New Delhi	SGPA: 9.90
CBSE (Class XII)	2025	Yuvashakti Model School	96.4 %
CBSE (Class X)	2023	Yuvashakti Model School	95.4 %

WORK EXPERIENCE

AI/ML Research Internship | Delhi Technological University

June 2026

- Conducting research on complexity-reduction strategies for the **ChARM architecture**, with emphasis on scalable temporal and inter-channel modeling in multimodal time-series learning.

AI/ML Research Internship | Netaji Subhas University of Technological

June 2026

- Researching on **context-aware object-centric surveillance**, with focus on open-vocabulary perception, long-term object memory, temporal reasoning, and scene-level relational understanding in video intelligence systems.

ACADEMIC PROJECT

Andrew Ng Digital Twin - Persistent Scientist AI System | [GitHub](#) | [Live Demo](#) | [Video Demo](#) | July 2026

- Deployed a role-adaptive Andrew Ng digital twin grounded in **529** sources, delivering citation-backed multi-turn conversations with **21/21 completed answers** passing deterministic persona and response-structure checks.
- Built timeline-aware graph memory, both in and cross-session recall, a live memory dashboard and streaming synthetic cloned voice; validated **93.3% of 15** memory scenarios, **95.2% of 21** assertions, **30/30** completed production streams and **7/7** security checks.

Octave - Context-Aware Spatial Interface | [GitHub](#) | [VideoDemo](#) |

May 2026

- Developed a multimodal Python/C++ interface combining offline NLP+CV, engineering dynamic context routing to translate **8** base gesture families into **15+** actions with **<5ms IPC** and **<80ms** end-to-end latency.
- Engineered a React/Electron dashboard for live retraining via **Transfer Learning** without catastrophic forgetting, hot-swapping PyTorch LSTMs/FFNNs (**~99%** static on **2,100 samples**, **~95.8%** dynamic) with **zero downtime**.

Bandhu.AI - Voice-Driven Universal Digital Inclusion System | [GitHub](#) | [Video Demo](#) |

April 2026

- Architected an interactive voice-to-web agent bridging **100M+ rural citizens** to government portals, processing DTMF and **22+ dialects** with **<3s response** and **<150ms API** latency to map unstructured audio into stateful JSON.
- Engineered an autonomous backend pipeline (FastAPI, Playwright) parsing JSON to submit **20+ dynamic** form fields in **<45 seconds**, reducing registration time from a **3-hour physical process** to a **5-minute automated call**.

Gesture-Controlled Drone System | [GitHub](#) | [Video Demo](#) |

January 2026

- Developed an edge-optimized, multimodal drone control interface fusing gesture, voice, and facial recognition (smile-based speed modulation), processing **7 distinct** hand gestures at **~30 FPS**.
- Implemented temporal smoothing to eliminate prediction jitter, achieving **~36ms end-to-end** latency and **~93% accuracy** via a custom ANN/CNN architecture.

ACADEMIC ACHIEVEMENTS AND AWARDS

- Academic Rank: 2nd** in Software Engineering Department (**9.9 SGPA**).
- Top 10 Participant (PS11):** Nationally in **ISRO Bhartiya Antariksh Hackathon 2026** for making PT-JEPA.
- Top 30 Participant:** National-Level [SahAI for Shiksha Hackathon 2026](#) by Wadhvani AI & Google.org.
- Top 10 Participant:** [KRITI Social Impact Challenge '26](#) at BITS Pilani (APOGEE 2026).
- JEE Main 2025 : 99+ Percentile(Top 1%)** nationwide (**AIR 14182 among 1M+ candidates**).

TECHNICAL SKILLS

- Languages:** C, C++, Python, SQL
- AI/ML:** PyTorch, Scikit-learn, Statsmodels, OpenCV, MediaPipe, Pandas, NumPy, RAG, STT, Chatterbox TTS.
- Tools & Frameworks:** Git/GitHub, FastAPI, React, Hugging Face, Playwright, Electron, Docker, PyPI, Vercel.
- Certificates:** [DL Specialization](#), [Sequence Models](#), [Convolutional Neural Networks](#), [ML Specialization](#), [Basics of Quantum Information](#)

POSITIONS OF RESPONSIBILITY

- Member, AIMS-DTU**